# Circle — Definition, Forms, Reality Check

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

LESSON 01 / 07

## Definition: fixed distance from a centre

A circle with centre $C=(h,k)$ and radius $r\ge 0$ is the locus

$$(x-h)^2+(y-k)^2=r^2.$$

For $r>0$ this is a proper circle; $r=0$ is the single point $C$.

**SHORT DERIVATION**

**Complete squares in the general equation**

For $x^2+y^2+2gx+2fy+c=0$, group the coordinate terms:

$$(x+g)^2+(y+f)^2=g^2+f^2-c.$$

Thus the centre is $(-g,-f)$ and $r^2=g^2+f^2-c$. The equal coefficients of $x^2, y^2$, with no $xy$ term, are essential for a circle in orthogonal Cartesian axes.

## Three cases—do not call every equation a circle

- $g^2+f^2-c>0$: a real circle of positive radius.
- $g^2+f^2-c=0$: a point-circle (one real point).
- $g^2+f^2-c<0$: no real points.

**WORKED EXAMPLE**

**Centre, radius, then check**

$x^2+y^2-2x-4y-4=0$ gives $g=-1, f=-2, c=-4$, centre $(1,2)$ and $r^2=1+4+4=9$, so $r=3$. Check $(1,5)$: it is distance $3$ from $(1,2)$ and satisfies the equation.

**COMMON TRAP**

**Radius is the nonnegative square root**

If a formula yields $r^2<0$, it is not a real circle. If $r^2=0$, do not describe it as a circle with positive circumference. A negative value cannot be repaired by dropping a sign.

LESSON 02 / 07

## Three independent conditions determine a circle

A general circle has equation $x^2+y^2+Dx+Ey+F=0$ with three unknown coefficients. Substituting three noncollinear points gives a linear system for $D, E, F$. If the points are collinear, no proper circle passes through all three.

**WORKED EXAMPLE**

**Through the origin and two unit intercepts**

Through $(0,0), (1,0), (0,1)$: the origin gives $F=0$, then $1+D=0$ and $1+E=0$. The circle is $x^2+y^2-x-y=0$, centre $(1/2, 1/2)$, radius $1/\sqrt{2}$.

## Diameter endpoints give a dot-product equation

For diameter endpoints $A=(x_1,y_1)$ and $B=(x_2,y_2)$, a point $P=(x,y)$ lies on the circle exactly when $\angle APB=90^\circ$ (Thales' theorem). Therefore

$$(P-A)\cdot(P-B)=0,$$

or explicitly

$$(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0.$$

The centre is the midpoint of $AB$ and the radius is $AB/2$.

**WORKED EXAMPLE**

**Diameter through two points**

For $A=(1,2)$, $B=(5,8)$, $(x-1)(x-5)+(y-2)(y-8)=0$, i.e. $(x-3)^2+(y-5)^2=13$. Both endpoints satisfy the diameter equation.

## Parametric equations

For centre $(h,k)$ and radius $r$,

$$x=h+r\cos t,\qquad y=k+r\sin t,\qquad 0\le t<2\pi.$$

# Determine a Circle and Parametrize It (cont.)

The angle parameter traces every point once except the equivalent endpoint at $2\pi$.

A rational parameter is

$$x = h + r\frac{1-u^2}{1+u^2}, \qquad y = k + r\frac{2u}{1+u^2}, \qquad u \in \mathbb{R}.$$

It traces the circle except $(h-r, k)$; that missing point is the limit $|u| \to \infty$. Include it separately when a problem asks for every point.

**COMMON TRAP**

**Parameter is not arc length**

In the trigonometric parametrization, $t$ is the polar angle about the centre. In the rational form, $u = \tan(t/2)$ and the omitted point comes from the half-angle coordinate's pole.

LESSON 03 / 07

## Centre-to-line distance classifies the intersection

Let a circle have centre $C = (h, k)$, radius $r$, and let the line be $ax + by + c = 0$. The distance from $C$ to the line is

$$d = \frac{|ah + bk + c|}{\sqrt{a^2 + b^2}}.$$

- $d < r$: two intersection points (a secant).
- $d = r$: one repeated intersection (a tangent).
- $d > r$: no real intersection.

**SHORT DERIVATION**

**Why the chord length is twice a right-triangle leg**

The perpendicular from the centre bisects a chord. If its length is $\ell$, the right triangle has hypotenuse $r$ and one leg $d$, hence $(\ell/2)^2 = r^2 - d^2$ and

$$\ell = 2\sqrt{r^2 - d^2}.$$

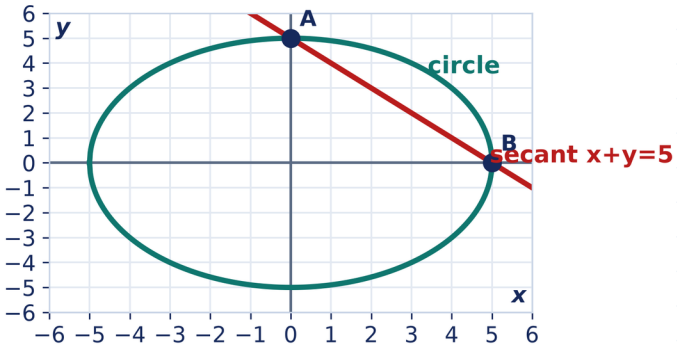
This is real precisely in the secant/tangent cases $d \le r$.

**WORKED EXAMPLE**

**A secant, not merely a line equation**

The line $x + y = 5$ is at distance $5/\sqrt{2}$ from the centre $(0, 0)$ of $x^2 + y^2 = 25$. Since this is less than $5$, it cuts a chord of length $2\sqrt{25 - 25/2} = 5\sqrt{2}$.

**Centre-to-line distance sets the chord length**




## Solve the intersection only after classifying it

Parametrize the line by $P(t) = P_0 + t(-b, a)$, where $P_0$ is any point on it. Substitute in $(x-h)^2 + (y-k)^2 = r^2$: a quadratic in $t$ results. Its discriminant is positive, zero or negative exactly when $d < r$, $d = r$ or $d > r$.

**COMMON TRAP**

**Tangency is a repeated root**

A tangent line meets the circle in one geometric point but gives a double root after substitution. Do not infer "no intersection" from a zero discriminant.

LESSON 04 / 07

## Tangent: perpendicular to the radius

For $(x-h)^2 + (y-k)^2 = r^2$ and a point $T = (x_1, y_1)$ on the circle, the tangent is

$$(x_1 - h)(x - h) + (y_1 - k)(y - k) = r^2.$$

The normal is the line through $T$ and the centre $(h, k)$.

# Tangent and Normal at a Point (cont.)

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

**SHORT DERIVATION**

***The tangent equation is a dot-product condition***

The radius vector $T - C$ is normal to the tangent. Thus $X$ is on the tangent exactly when $(T - C) \cdot (X - T) = 0$. Rearranging gives $(T - C) \cdot (X - C) = |T - C|^2 = r^2$.

For $x^2 + y^2 + 2gx + 2fy + c = 0$, the same tangent is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0.$$

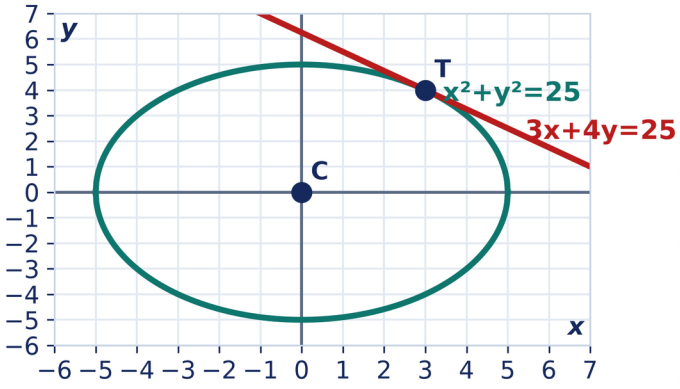
This formula assumes $T$ lies on the circle; the "$T = 0$" replacement is a mnemonic, not a substitute for checking incidence.

**WORKED EXAMPLE**

***Unit-free tangent check***

For $x^2 + y^2 = 25$ at $T = (3, 4)$, the tangent is $3x + 4y = 25$. It contains $T$, and its normal vector $(3, 4)$ is the radius direction.

**The radius is normal to the tangent**




**COMMON TRAP**

***A tangent is attached to a point on the circle***

If a proposed point is not on the circle, the at-point tangent formula does not apply. From an external point there are generally two tangents; derive them as lines through that point with discriminant zero or use the polar/chord-of-contact equation.

LESSON 05 / 07

## Power records inside, on, or outside

For circle centre $C = (h, k)$ and radius $r$, define

$$\text{Pow}(P) = |P - C|^2 - r^2.$$

It is negative inside, zero on, and positive outside. In the general equation $x^2 + y^2 + 2gx + 2fy + c = 0$, the power at $(x_0, y_0)$ is the left-hand side evaluated there.

**SHORT DERIVATION**

***Tangent length from a right triangle***

If $P$ is outside and $T$ is a tangent point, $CT \perp PT$. Therefore

$$PT^2 = PC^2 - CT^2 = PC^2 - r^2 = \text{Pow}(P).$$

So each tangent from $P$ has length $\sqrt{\text{Pow}(P)}$.

**WORKED EXAMPLE**

***Tangent length is not centre distance***

From $P = (5, 5)$ to $x^2 + y^2 = 9$, power is $25 + 25 - 9 = 41$. Each tangent has length $\sqrt{41}$, whereas $PC = \sqrt{50}$.

## Chord of contact / polar of an external point

For a circle centred at the origin, $x^2 + y^2 = r^2$, the two tangent points from $P = (p, q)$ lie on the line

$$px + qy = r^2.$$

This line is the ***chord of contact***. Its intersection with the circle gives the two contact points. For a translated circle, first use coordinates relative to its centre: $(P - C) \cdot (X - C) = r^2$.

**WORKED EXAMPLE**

***Find both contacts by one line and one circle***

From $P = (5, 0)$ to $x^2 + y^2 = 9$, the chord of contact is $5x = 9$, so $x = 9/5$. Then $y^2 = 9 - 81/25 = 144/25$, giving contacts $(9/5, \pm 12/5)$. The tangent length is $\sqrt{25 - 9} = 4$.

COMMON TRAP

**The polar formula needs the same circle centre**

For a circle not centred at the origin, replace $P$ and $X$ by their vectors from the centre. Forgetting this translation shifts the chord of contact.

LESSON 06 / 07

## Subtract normalized circle equations

Write $S_i = x^2 + y^2 + D_i x + E_i y + F_i$ for $i = 1, 2$; the corresponding circles are $S_i = 0$. Their power functions are $S_1(x, y)$ and $S_2(x, y)$. Equal power means $S_1 - S_2 = 0$; the quadratic terms cancel, leaving a line (unless the circles are concentric or identical).

SHORT DERIVATION

**Why the line is perpendicular to the centres**

The centre difference is parallel to $(D_1 - D_2, E_1 - E_2)$, the normal vector of $S_1 - S_2 = 0$. Thus the radical axis is perpendicular to the line of centres. When the real circles intersect, its points are exactly their common chord.

WORKED EXAMPLE

**Subtract, then test equal powers**

For $x^2 + y^2 = 9$ and $x^2 + y^2 - 4x + 6y - 3 = 0$, subtract to get $2x - 3y - 3 = 0$. At $(0, -1)$ both power values equal $-8$, checking the line's equal-power meaning.

## Three circles and the radical centre

The radical axes of three circles are the pairwise equal-power lines. If the circles are not identical and the axes are not parallel, solve two axes to get their common point; equality of the third pair follows by subtraction. Degenerate cases need care: concentric unequal circles have no finite equal-power locus, identical circles have equal power everywhere, and parallel axes need not meet in a finite radical centre.

COMMON TRAP

**A radical axis need not be a real common chord**

Two nonconcentric circles can be disjoint or nested. Their equal-power line still exists, but it contains real points only as a locus of equal powers—not points on both circles. Call it a common chord only when the circles actually intersect.

LESSON 07 / 07

## Compare centre distance and radii

For centres distance $d$ and radii $r_1, r_2 > 0$:

| Condition | Real intersection |
|---|---|
| $d > r_1 + r_2$ | none; externally separate |
| $d = r_1 + r_2$ | one; externally tangent |
| $\lvert r_1 - r_2 \rvert < d < r_1 + r_2$ | two points |
| $d = \lvert r_1 - r_2 \rvert > 0$ | one; internally tangent |
| $0 < d < \lvert r_1 - r_2 \rvert$ | none; one strictly inside the other |
| $d = 0$ | concentric: compare radii; identical if radii agree |

The zero-radius point-circle is handled separately by testing whether its point lies on the other circle.

# Efficient order of attack

## Efficient order of attack

1. Complete squares and identify centres/radii.
2. For a line, use centre-to-line distance before solving intersections.
3. For a tangent, check the point lies on the circle and use the radius as normal.
4. For a common chord, subtract normalized equations first.
5. For two circles, compare $d$ with $r_1 + r_2$ and $|r_1 - r_2|$.

**WORKED EXAMPLE**

***Two equal circles***

$(x-0)^2 + y^2 = 4$ and $(x-3)^2 + y^2 = 4$ have $d = 3$, $r_1 + r_2 = 4$, and $|r_1 - r_2| = 0$. Hence they meet twice. Subtracting gives radical axis $x = 3/2$; substituting yields $y = \pm\sqrt{7}/2$.

**ASSUMPTIONS & CONVENTIONS**

***JEE boundary***

The official circle bullet includes circle forms, tangent, normal, chord, parametric equations and intersection with a straight line. Radical axes, power and two-circle position are useful direct applications; the official scope is not a promise to include every theorem about circles.